Highlights

Semantic Similarity Detection in Theses Using Fine-Tuned IndoBERT

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- Fine-tuned IndoBERT models semantic similarity in thesis literature reviews.
- Forty-five Chapter II documents yielded over 5,000 pairs per scenario.
- Five raters showed high agreement with the automatic similarity labels.
- Pearson correlation reached 0.9605 under document-based 5-fold CV.
- Errors clustered near the 0.66 and 0.88 similarity thresholds.

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ABSTRACT

The growing accessibility of digital references has heightened the risk of plagiarism in undergraduate thesis literature reviews, particularly through paraphrasing and sentence-structure alteration that lexical matching methods may miss. This study develops a semantic similarity detection system based on a fine-tuned IndoBERT Siamese Bi-Encoder for long-form Indonesian academic texts. The system was constructed from 45 thesis Chapter II documents and evaluated under three preprocessing scenarios, S1, S1mc, and S1mca, yielding 5,189, 5,065, and 5,044 sentence pairs, respectively. Similarity scores were computed directly from cosine similarity and evaluated using document-based 5-Fold GroupKFold cross-validation. Human annotation of 359 sample pairs by five raters achieved a Krippendorff's Alpha of 0.8070 and ICC(A,k) of 0.9548, while automatic labels correlated with mean annotator scores at $r = 0.9804$. S1mca produced the highest Pearson correlation (0.9605 ± 0.0031), whereas S1 yielded the lowest RMSE (0.0925) and MAE (0.0661). Four-zone accuracy reached 99.4-99.8% in the high zone and 80.4-82.4% in the negative zone, but was lower in the medium and low zones. Dominant errors were medium-to-high and low-to-medium zone shifts.

Introduction

Advances in information technology and the increasingly broad accessibility of digital literature have transformed the process of academic writing in higher education; however, this ease of access has concurrently heightened the risk of plagiarism when not accompanied by adequate understanding of academic ethics (Amalina and Ardiansyah, 2025). Chapter II (Literature Review) is the section most susceptible to plagiarism, as it comprises a compilation of theories, definitions, and prior research findings that inherently exhibit a degree of inter-document similarity; a study of 79 undergraduate thesis proposals recorded an average Chapter II similarity index of 71% (Oktapiani, Sutiono, Rodhiyana, Fadilatus and Salim, 2023). This vulnerability is further compounded by students' limited understanding of appropriate paraphrasing techniques and source attribution (Budiman and Widjaja, 2020), constrained ability to construct a coherent and original literature review (Risparyanto, 2020), as well as time pressure and reliance on insufficiently verified online sources.

Plagiarism detection systems commonly employed in academic settings continue to rely predominantly on lexical matching approaches such as n-gram and string-based similarity. Although effective for literal plagiarism, such approaches are limited in detecting paraphrasing, sentence-structure alteration, and synonym substitution, as they are inadequately equipped to capture semantic equivalence between texts that are meaningfully similar yet lexically divergent (Clough and Stevenson, 2011). The development of transformer-based language models such as BERT has enabled semantic similarity approaches that focus on meaning and context, demonstrating superior performance on semantic textual similarity tasks relative to traditional methods (Reimers and Gurevych, 2019). Within the Indonesian-language context, IndoBERT, as a dedicated pretrained model, has been employed in short-sentence paraphrase detection (Kartika, Alfredo and Kusuma, 2023; Wilie, Vincentio, Winata, Cahyawijaya, Li, Lim, Soleman, Mahendra, Fung, Bahar and Purwarianti, 2020) and in the classification of plagiarism patterns in thesis documents. However, these studies generally confine their scope to short texts or categorical classification outputs, while validation of label quality through explicit human annotation reliability remains rarely examined. Siamese architecture-based studies for continuous semantic similarity measurement in long-form Indonesian academic documents, accompanied by annotation reliability validation, thus remain an underexplored gap.

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Related work and research gap

This section discusses prior studies relevant to semantic similarity-based plagiarism detection, transformer-based language models, Siamese bi-encoder architectures, and annotation reliability validation. The discussion concludes with the identification of the research gap underlying the contribution of this study.

Plagiarism detection aims to identify similarities between two or more text documents, either as direct verbatim copying or as obfuscation through paraphrasing and changes in sentence structure. Foltýnek, Meuschke and Gipp (2020) conducted a systematic literature review of academic plagiarism detection methods and concluded that string-matching approaches still dominate widely used systems, but consistently fail to recognize plagiarism disguised through paraphrasing, synonym replacement, and sentence restructuring because similarity is measured at the level of word occurrence rather than meaning. Similar findings were reported by Rosu, Stoica, Popescu and Mihaescu (2020), who proposed a deep-learning approach and showed that neural-network-based semantic representations improved detection capability compared with conventional rule-based and string-similarity methods.

In the Indonesian academic context, Shadiqi (2019) examined factors that encourage plagiarism in academic writing, including limited understanding of appropriate paraphrasing and source attribution and difficulties in constructing a coherent and original literature review. The Literature Review section of undergraduate theses is particularly vulnerable to document similarity because it summarizes theories, definitions, and previous research findings from a relatively limited set of sources that often overlap among students in the same study program.

The transformer architecture introduced by Vaswani et al. (Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser and Polosukhin, 2017) replaced recurrence mechanisms with self-attention, enabling each token to attend to all other tokens within a sentence in parallel. Building on this architecture, Devlin, Chang, Lee and Toutanova (2019) developed BERT, trained through two pretraining tasks, namely Masked Language Modeling and Next Sentence Prediction, thereby producing bidirectional contextual representations that outperform static embedding-based models and earlier recurrent architectures across a range of natural language understanding tasks.

Xian, Yuan, Zheng, Chen and Yuntao (2024) implemented BERT as a matching engine in a coursework plagiarism detection system and reported that BERT-based representations were more effective at recognizing semantic similarity than fingerprint- and n-gram-based matching methods, particularly in paraphrase cases. Fatwanto, Zamakhsyari, and Ndungi (Fatwanto, Zamakhsyari and Ndungi, 2024) found that BERT variants trained specifically on a given language consistently outperformed multilingual models on tasks within that language, as the resulting representations more closely align with the target language's morphological and lexical characteristics.

Within the Indonesian-language context, Koto, Rahimi, Lau and Baldwin (2020) developed IndoBERT as part of the IndoLEM benchmark and demonstrated consistent performance gains over mBERT on sentiment analysis, named entity recognition, and textual similarity tasks. Wilie et al. (Wilie et al., 2020) extended this evaluation through the IndoNLU benchmark, encompassing twelve Indonesian-language NLU tasks, and released an IndoBERT variant trained on a corpus with broader domain coverage. These two studies form the basis for the selection of IndoBERT as the pretrained encoder in this study.

Although standard BERT excels at sentence-pair classification tasks through a cross-encoder scheme, this approach is inefficient for large-scale similarity comparison, as each text pair must be jointly reprocessed. Reimers and Gurevych (Reimers and Gurevych, 2019) proposed Sentence-BERT, a modification of the BERT architecture employing a Siamese and triplet network structure with a pooling operation, enabling each sentence to be independently mapped into a fixed-dimensional vector space. This approach reduced the time required to find the most similar sentence pair from tens of hours to a matter of seconds, without meaningful degradation in accuracy on Semantic Textual Similarity tasks.

Sarkar, Sarkar, Ansar and Chakrabarti (2026) applied a Siamese BERT architecture to evaluate semantic similarity between research papers and demonstrated that this architecture effectively captures topical similarity in scientific documents ranging from paragraph to full-document scale, although their study focused on English-language text and did not incorporate explicit annotation label reliability validation through human raters.

The quality of reference labels is a determining factor in the reliability of semantic similarity models, particularly when labels are constructed through human annotation or with the assistance of large language models. Wang, Kim, Rahman, Mitra and Miao (2024) found that human verification of LLM-generated labels remains necessary to preserve dataset validity, as unverified labels risk propagating systematic bias into models trained on them. Wang, Tao, Xie, Yang, Baldwin and Verspoor (2023) demonstrated that aggregating scores across multiple raters produces more representative labels than single-rater scores, consistent with the findings of Guo et al. (2017) in the construction of the CORD19STS dataset..

The literature synthesis reveals four relevant gaps. First, lexical approaches are inadequate for paraphrasing and sentence restructuring (Foltýnek et al., 2020). Second, IndoBERT has been extensively evaluated on NLU and paraphrase detection tasks, yet evidence of its application to similarity score regression on long-form Chapter II documents remains limited (Kartika et al., 2023; Koto et al., 2020; Wilie et al., 2020). Third, Siamese/Bi-Encoder architectures offer efficiency for large-scale comparison, but validation within the Indonesian academic domain remains limited (Reimers and Gurevych, 2019; Sarkar et al., 2026). Fourth, the quality of automatically generated labels and the risk of data leakage have not consistently been treated as a core component of experimental design. This study addresses these gaps by integrating a fine-tuned IndoBERT model, a Siamese Bi-Encoder architecture, five-annotator validation, and document-based 5-Fold GroupKFold cross-validation within a single experimental framework. Table 1 summarizes how this study positions itself relative to these prior studies.

Table 1: Comparison of prior research gaps with the proposed contribution.

Study	Method	Scope/Output	Document-Based CV
Xian et al. (2024)	SBERT, FAISS, fine-tuned BERT (retrieval)	Short text, non-Indonesian	No
Kartika et al. (2023)	Fine-tuned IndoBERT with GPT augmentation	Short sentences, Indonesian	No
Sarkar et al. (2026)	Siamese BERT	Research papers, non-Indonesian	No
This study	Fine-tuned IndoBERT Siamese Bi-Encoder with 3 masking schemes	Continuous scores, long-form Indonesian documents	Yes (5-fold GroupKFold)

In light of these issues, this study addresses how a semantic similarity detection system can be implemented and evaluated for Chapter II of Indonesian-language undergraduate theses through a fine-tuned IndoBERT model built on a Siamese Bi-Encoder architecture. This study aims to (1) develop and evaluate a semantic similarity detection model for Chapter II of Indonesian-language undergraduate theses using a fine-tuned IndoBERT model trained on an academic text dataset within a Siamese Bi-Encoder architecture, and (2) objectively and measurably evaluate the model's performance using a document-based 5-Fold Cross-validation scheme with Pearson correlation, RMSE, MAE, and four-class zone accuracy metrics. This system is expected to address the limitations of lexical matching methods and to contribute to the preservation of academic integrity within higher education institutions.

Materials and Methods

This study constructs and evaluates a semantic similarity detection model for Chapter II of Indonesian-language undergraduate theses using a Fine-Tuned IndoBERT Siamese Bi-Encoder approach. The research stages comprise data preparation and characterization, text preprocessing, construction of the text-pair dataset together with label quality validation, model architecture design and optimization strategy, and a cross-validation-based evaluation scheme.

Data source and characteristics

This study uses a dataset consisting of 45 Chapter II (Literature Review) documents from undergraduate theses of students in the Undergraduate Informatics Study Program, Telkom University Surabaya, in PDF format. Documents were selected based on three criteria: the text could be extracted in full, the text was written in Indonesian, and the document originated from the Informatics study program, ensuring that the dataset was reasonably homogeneous in disciplinary domain while still representing variation in writing style across students.

All documents were used solely for the purposes of this study. Prior to processing, each document was anonymized by removing student names, student identification numbers (NIM), supervisor names, and all other identifying metadata from the cover page and approval sheet, such that only the content of Chapter II was used in the extraction and model training stages.

Research workflow

The research workflow was designed as a sequence of ordered stages, as illustrated in Figure 1, the workflow comprises text extraction and preprocessing, construction of labeled text pairs, label quality validation, model fine-tuning, and cross-validation-based evaluation. The extraction stage retrieved the content of Chapter II from each PDF document using the PyMuPDF library, with chapter boundaries identified through regular expression pattern matching ("BAB II" through the point preceding "BAB III") and paragraph structure detected based on vertical line spacing (Y-gap) and bounding-box position, in order to separate narrative text from non-narrative elements such as tables, figure captions, and page numbers. Extraction quality was verified through manual inspection of a 10% document sample.

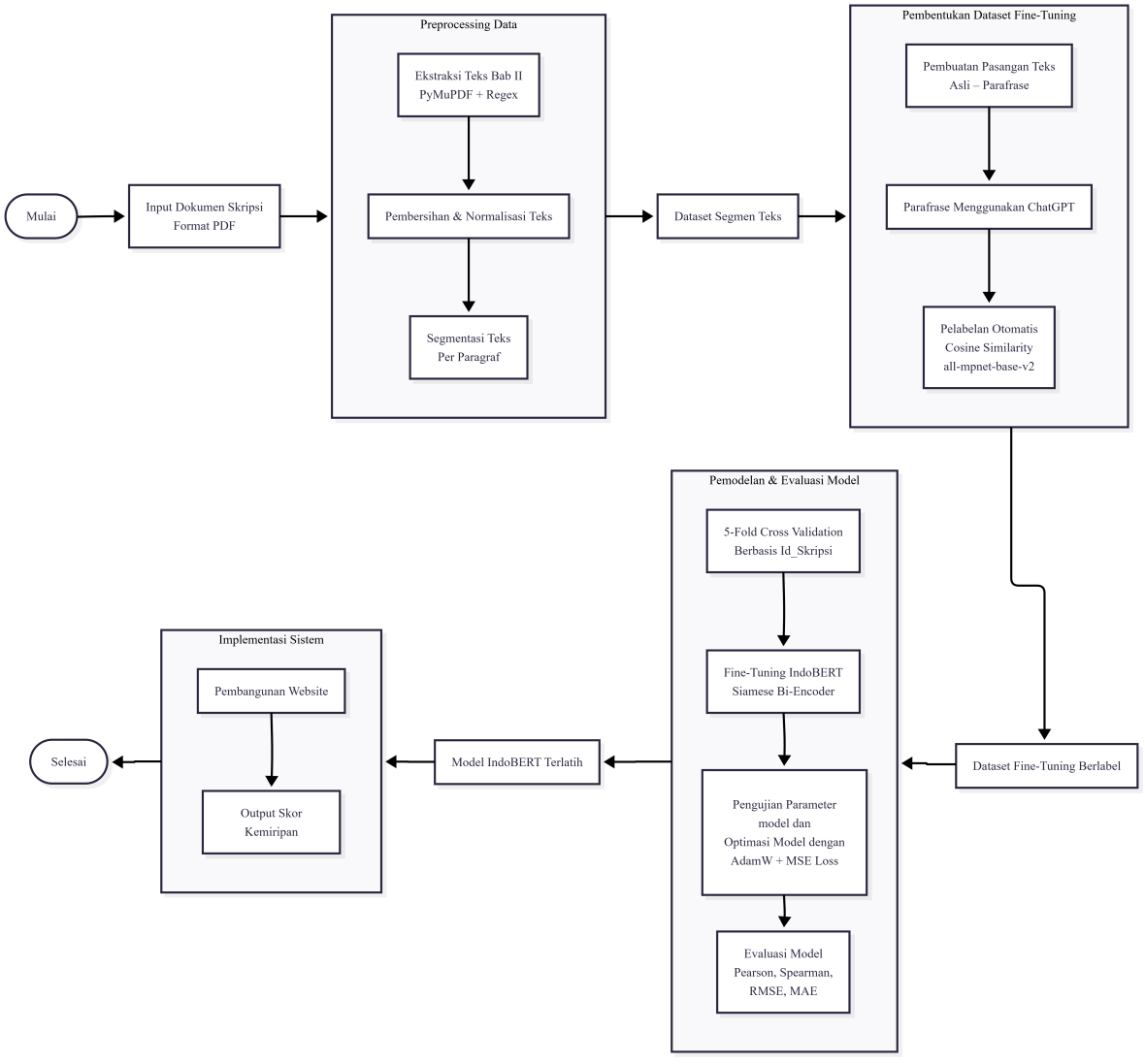


Figure 1: Research workflow.

The extracted text subsequently underwent a cleaning stage, comprising the removal of non-informative elements (page numbers, non-standard symbols, double spacing) and normalization of Unicode characters and punctuation, prior to segmentation using per-paragraph segmentation combined with a sliding-window mechanism based on the IndoBERT tokenizer (maximum length of 300 tokens, overlap of 150 tokens). Segments shorter than 20 words were discarded, and deduplication was performed using SHA-256 hashing on the normalized text. This process yielded 5,189 pairs under the S1 scenario, 5,065 pairs under S1mc, and 5,044 pairs under S1mca, all derived from the same 45 documents, with no missing values or out-of-range labels.

Construction of the text-pair dataset

Text pairs for training were constructed from Chapter II segments using a combination of automatic paraphrasing and cross-paragraph pairing, subsequently grouped into four similarity zones: high, medium, low, and negative. The high and medium zones were obtained through direct paraphrasing using GPT-4o-mini at varying degrees of alteration, whereas the low and negative zones were constructed by pairing segments from different subsections or documents without a paraphrasing process. Initial labels for each pair were obtained using paraphrase-multilingual-mpnet-base-v2 through the computation of cosine similarity between embeddings, with Word Change Rate (WCR) employed as an additional criterion to ensure an adequate degree of lexical alteration in paraphrased pairs. The zone boundaries are presented in Table 2.

Table 2: Similarity zone boundaries and characteristics.

Zone	Score Range	Characteristics
High	0.88–1.00	Identical meaning, limited synonym substitution, sentence structure largely preserved
Medium	0.66–0.88	Flexible structural alteration, core meaning preserved, differing viewpoint
Low	0.45–0.66	Substantial contextual difference, still within the same topic
Negative	< 0.45	No shared meaning (cross-paragraph, hard negative)

Automatic labeling was employed to construct a training dataset of adequate scale, as manual labeling of the entire pair set would require substantial annotation time and the involvement of annotators with sufficient domain understanding. This consideration is particularly relevant given that the dataset comprises thousands of text pairs spanning varying degrees of similarity, such that applying human annotation across the entire set would likely be inefficient and difficult to maintain consistently. This approach, however, carries a limitation, as labels generated through cosine similarity reflect characteristics of the source encoder's embedding space and do not fully constitute a measure of semantic similarity free of model-induced bias. Furthermore, differences in pair-construction strategy between paraphrase-derived data and cross-paragraph pairing may introduce surface-level cues that the model could learn as spurious zone-discriminating signals.

To independently examine the validity of the automatic labels, human validation was conducted on 359 pairs selected via Stratified Random Sampling based on zone distribution, with sample size determined using Cochran's formula at a 95% confidence level and a 5% margin of error. Each pair was independently rated by five students from the Informatics Study Program, Telkom University Surabaya, using a continuous 0-1 scale, where 0 indicates a pair that is highly dissimilar in meaning and 1 indicates a pair that is highly similar or nearly identical in meaning. Annotators were not provided with information regarding the four zones, the cosine similarity boundaries, the automatic labels, or the pair-construction strategy, such that ratings were based entirely on perceived semantic similarity. An annotation guideline explaining the rating procedure and criteria is provided as supplementary material. The five annotators' ratings were subsequently used to measure inter-annotator reliability using Krippendorff's Alpha and the Intraclass Correlation Coefficient (ICC), while the mean score across the five annotators was used as the human reference label, rather than as an absolute ground truth. This reference label was applied to the validation subset to examine the relationship between model predictions and independent human semantic judgment, separate from the automatic labels, thereby enabling the evaluation to distinguish the model's ability to follow the automatic label pattern during training from its ability to represent human perception of semantic similarity.

Model architecture and fine-tuning strategy

The model was built using a Siamese Bi-Encoder architecture with an IndoBERT backbone (indobenchmark/indobert-base-p2). The two text segments within a given pair were processed in parallel through the same encoder (shared weights), after which the token-level outputs were aggregated using masked mean pooling and normalized via L2 normalization, as illustrated in Figure 2 and formulated in Equations (1) and (2).

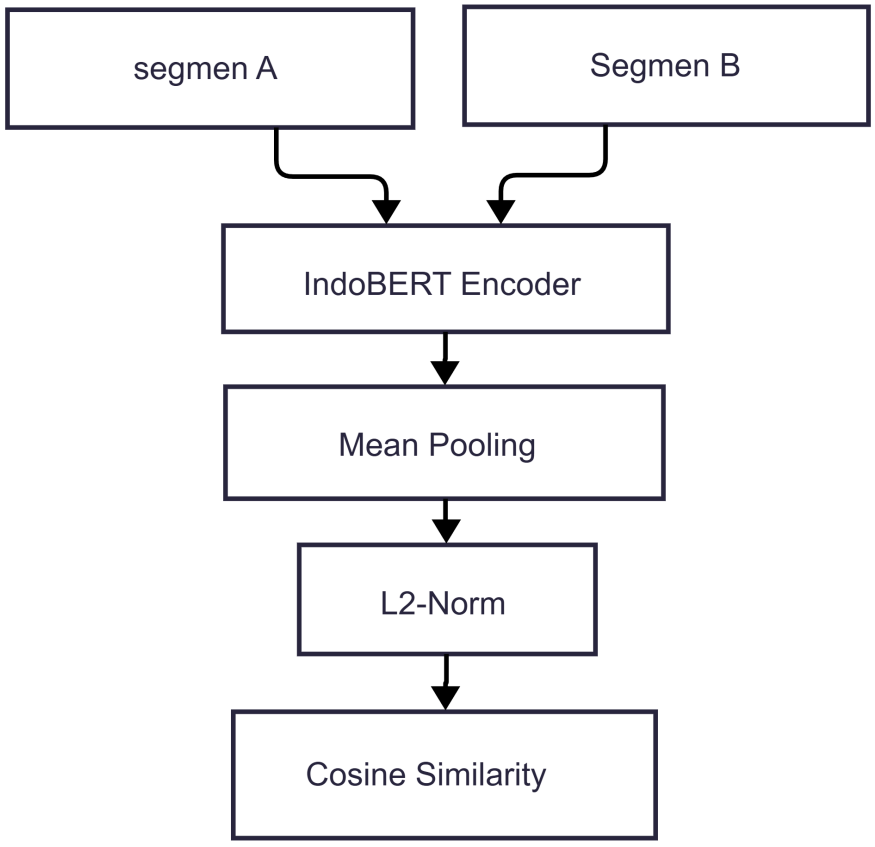


Figure 2: Implementation of the Siamese Bi-Encoder architecture.

$$e = \frac{\sum_{i=1}^{n} m_i h_i}{\sum_{i=1}^{n} m_i}. \tag{1}$$

$$\hat{e} = \frac{e}{\|e\|_2}. \tag{2}$$

The predicted similarity score was computed directly as the cosine similarity between the two normalized embedding vectors, as shown in Equation (3), without additional range transformation, such that the model output is directly comparable to the continuous label.

$$s = \text{cosine_similarity}(u, v) = \frac{u \cdot v}{\|u\| \, \|v\|}. \tag{3}$$

The training process employed the Mean Squared Error (MSE) loss function, as shown in Equation (4), against the continuous label.

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{j=1}^{N} (y_j - \hat{y}_j)^2. \tag{4}$$

Backpropagation used the AdamW optimizer with a weight decay of 0.01, accompanied by a linear learning-rate scheduler with a warmup ratio of 0.1, and early stopping based on Pearson correlation on the validation data, with a

patience of 3 epochs and a maximum limit of 10 epochs per fold. To assess the influence of non-semantic elements on model performance, training and testing were conducted across three text preprocessing scenarios: S1 (raw text without modification), S1mc (masking of academic citations), and S1mca (masking of citations and visual references such as figures, tables, and equations). All three scenarios were trained and evaluated separately using an identical architecture configuration and optimization strategy.

Evaluation scheme and metrics

Model performance was evaluated using a document-based 5-Fold cross-validation scheme (GroupKFold), ensuring that all segments originating from the same thesis document were consistently assigned to the same fold, thereby preventing data leakage between training and testing data, as illustrated in Figure 3. Cross-validation-based evaluation provides a more accurate depiction of model performance on unseen data compared with a single random split (Arlot and Celisse, 2010). At each iteration, the model was trained on four folds and tested on the remaining fold, with results averaged across all five folds.

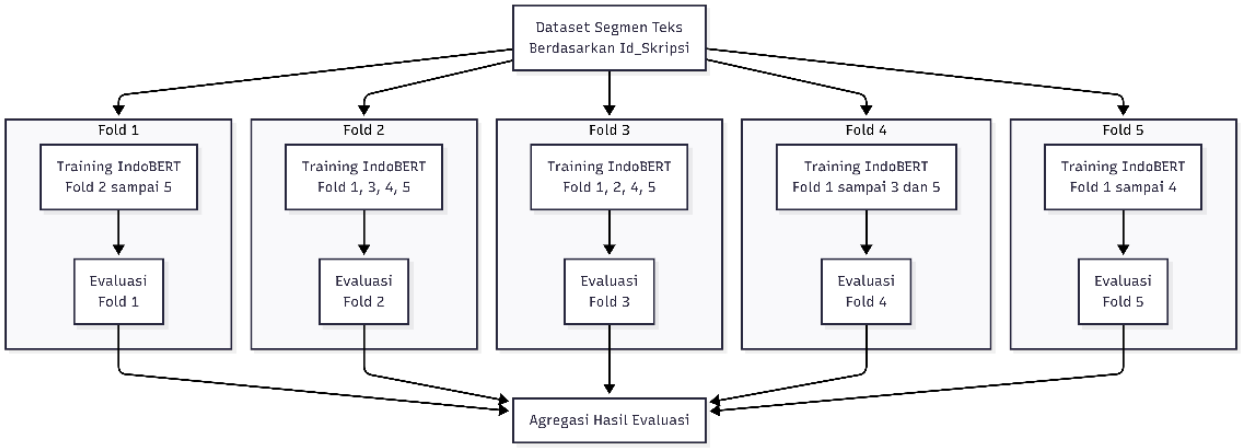


Figure 3: Document-based 5-Fold GroupKFold cross-validation scheme.

Model performance was measured using three regression metrics: Pearson correlation, to assess the linear relationship between predicted scores and actual labels; Root Mean Squared Error (RMSE), to assess error with greater emphasis on large deviations; and Mean Absolute Error (MAE), to assess average error with greater stability against outliers. In addition to regression evaluation, continuous prediction scores were converted into four similarity zones using the same thresholds applied during the labeling stage, namely the high zone for scores above 0.88, the medium zone for scores above 0.66, the low zone for scores above 0.45, and the negative zone for scores below 0.45. Zone evaluation was reported as four-class (exact-match) accuracy rather than binary accuracy, as binary accuracy at the 0.45 threshold would be uninformative for the high, medium, and negative zones, whose members lie entirely on one side of that threshold.

Comparative baselines

To assess the contribution of fine-tuning independently of the labeling process, the main model's performance was compared against four baseline methods under an identical document-based 5-Fold GroupKFold scheme: (i) TF-IDF (unigram-bigram, sublinear TF) with cosine similarity, fitted only on the training data of each fold; (ii) character n-gram (3-5) Jaccard and (iii) word n-gram (1-2) Jaccard, both requiring no training process; and (iv) IndoBERT-base-p2 without fine-tuning, using masked mean pooling and L2 normalization, with the same output scale as the main model (cosine similarity clipped to the [0,1] range). The multilingual Sentence-BERT model (paraphrase-multilingual-mpnet-base-v2) without fine-tuning was not included as a comparative baseline, as this was the very model used to generate the automatic labels during dataset construction (the Construction of the Text-Pair Dataset subsection); its output would therefore nearly reproduce the labels circularly and would not reflect genuine generalization capability. As direct evidence, when run under the exact configuration used at the labeling stage (maximum token length of 128), the zero-shot SBERT score perfectly reproduced label_cont (Pearson $r = 1.0000$; RMSE = MAE = 0.0000), confirming

that this model's high score is purely a consequence of procedural identity rather than an indication of genuine model capability.

Hardware and software specifications

All fine-tuning and evaluation experiments were executed in a consistent environment. The processor was an AMD Ryzen 5 6600H with 16 GB DDR5 RAM and an NVIDIA GeForce RTX 3050 with 4 GB VRAM. The operating system was Windows 11 Pro 64-bit, with Visual Studio Code and Google Colab used as development environments. Experiments used Python 3.10, PyTorch, SentenceTransformers, scikit-learn, NumPy, Pandas, Matplotlib, NLTK, transformers, and tqdm. All experiments used a fixed random seed of 42 across Python, NumPy, PyTorch, and CUDA for reproducibility.

Statistics

The significance of the performance difference between the main model and each baseline was tested using document-level cluster bootstrap (B = 2,000 resamples), as paired testing across folds (n = 5) has limited statistical power due to overlapping training data between folds. In each resample, the 45 documents were resampled with replacement, and all methods were evaluated on the same document sample, such that the differences between methods remained paired; a difference was considered significant when the 95% confidence interval did not include zero. As a complement, a paired t-test and Wilcoxon signed-rank test on the five fold-level values were also conducted for each main model-baseline pair: the paired t-test showed strong significance across all baselines and metrics ($p < 0.001$), while the Wilcoxon signed-rank test yielded $p = 0.0625$ consistently across all comparisons - the mathematically minimum attainable p-value for this test at $n = 5$ paired samples, and thus not an indication of a weak difference but rather a limitation of statistical power at this small fold sample size. The document-level cluster bootstrap ($n = 45$) was accordingly adopted as the primary test, given its substantially larger effective sample size; all differences under this test were likewise significant at an empirical level of p_boot approximately 0.001 (B = 2,000 resamples). Table 3 summarizes these three significance tests for the four baselines under the S1 scenario, allowing the magnitude, direction, and cross-test consistency of the results to be compared directly.

Table 3: Summary of significance tests for the main model versus baselines under the S1 scenario (Pearson metric).

Baseline	Δ Pearson (Main − Baseline)	95% Bootstrap CI (Document)	p_boot	p (paired t-test)	p (Wilcoxon signed-rank)
TF-IDF	+0.1130	[+0.1081; +0.1183]	≈ 0.001	< 0.001	0.0625
Character n-gram Jaccard (3–5)	+0.1461	[+0.1402; +0.1523]	≈ 0.001	< 0.001	0.0625
Word n-gram Jaccard (1–2)	+0.1760	[+0.1699; +0.1826]	≈ 0.001	< 0.001	0.0625
IndoBERT without fine-tuning	+0.0903	[+0.0812; +0.0998]	≈ 0.001	< 0.001	0.0625

The p_boot values for all baselines reach the empirical precision limit of $1/(B+1)$ approximately 0.001 at B = 2,000 resamples, and are accordingly reported as approximately 0.001 rather than $p = 0$. The Wilcoxon column is identical (0.0625) across all rows because this is the mathematically minimum attainable value for the signed-rank test at $n = 5$ paired folds, not an indication of coincidentally uniform results.

Results

Dataset quality validation

This section presents the experimental results of the Fine-Tuned IndoBERT Siamese Bi-Encoder model along with their discussion, following the stages outlined in the Method section: dataset quality validation, performance comparison across preprocessing scenarios, per-zone similarity performance evaluation, and discussion of the factors influencing model performance and its limitations.

The reliability of human annotation on the 359 sample pairs was measured using Krippendorff's Alpha and the Intraclass Correlation Coefficient (ICC), while the agreement between the automatic labels and the mean rating of the five annotators was measured using Pearson correlation, Spearman correlation, MAE, and RMSE. Table 4 summarizes the results of both assessments.

Table 4: Summary of dataset validation.

Dimension	Metric	Value
Inter-rater reliability	Krippendorff's Alpha	0.8070
Inter-rater reliability	ICC(A,k)	0.9548
Label vs. Annotator	Pearson r	0.9804
Label vs. Annotator	Spearman rho	0.9685
Label vs. Annotator	MAE / RMSE	0.1366 / 0.1464

The Krippendorff's Alpha value of 0.8070 and ICC(A,k) of 0.9548 indicate a high level of agreement among annotators, supporting the use of the five annotators' mean score as a human reference. The Pearson correlation of 0.9804 between the automatic labels and the mean annotator rating indicates a high degree of agreement, with a mean discrepancy of 0.1366 points on the 0-1 scale. The automatic labels were accordingly deemed suitable for use as ground truth in the fine-tuning process.

Performance comparison across preprocessing scenarios

The three scenarios were evaluated using a grid search over learning rate (2e-05 and 3e-05) and batch size (16 and 32) through document-based 5-Fold cross-validation. Table 5 summarizes the best configuration for each scenario, selected based on the mean Pearson correlation across folds.

Table 5: Best configuration for each preprocessing scenario.

Scenario	lr / batch	Pearson	Spearman	RMSE	MAE	F1-Macro
S1	2e-05 / 32	0.9602 ± 0.0020	0.9687	0.0925	0.0661	0.9004
S1mc	2e-05 / 32	0.9602 ± 0.0027	0.9688	0.0941	0.0672	0.8857
S1mca	2e-05 / 16	0.9605 ± 0.0031	0.9690	0.0941	0.0670	0.8893

The Pearson correlation difference across scenarios is only 0.0003 (0.9602 for S1 and S1mc, 0.9605 for S1mca), falling below the between-fold standard deviation of each scenario (0.0020 to 0.0031). This difference cannot be interpreted as indicating the superiority of any single masking strategy. On the error metrics, S1 in fact produced the lowest RMSE (0.0925) and MAE (0.0661), as well as the highest F1-Macro (0.9004), indicating that citation and visual-reference masking did not yield a measurable performance improvement on this dataset.

A learning rate of 2e-05 was selected across all three scenarios, whereas the optimal batch size differed between 32 (S1, S1mc) and 16 (S1mca). For S1mca, the combination of lr 3e-05 and batch size 32 produced a lower RMSE (0.0913) and higher F1-Macro (0.8969) than the selected configuration, indicating that the configuration ranking depends on the selection metric used. This variation in optimal batch size across scenarios is consistent with findings that batch size influences model generalization through the shape of minima on the loss surface, with large batches tending to converge to sharp minima that generalize more poorly (Keskar, Mudigere, Nocedal, Smelyanskiy and Tang, 2017), while small batches tend to produce flatter minima that are more stable on test data (Masters and Luschi, 2018).

To statistically test the significance of these differences, a paired Wilcoxon signed-rank test was conducted on the five-fold Pearson correlations for the best configuration of each scenario. The results showed no significant difference across any scenario pair (S1 vs. S1mc: $p = 0.8125$; S1 vs. S1mca: $p = 0.6250$; S1mc vs. S1mca: $p = 0.6250$; $n = 5$

folds), consistent with the paired t-test results for all three pairs ($p = 0.9983$; $p = 0.6453$; $p = 0.9137$). With $n = 5$ folds, the statistical power of this test is limited; however, the direction and magnitude of the mean Pearson differences across scenarios (0.00000 to 0.00028), substantially smaller than the between-fold standard deviation, reinforce the conclusion that citation and visual-reference masking do not demonstrate a statistically meaningful difference in performance.

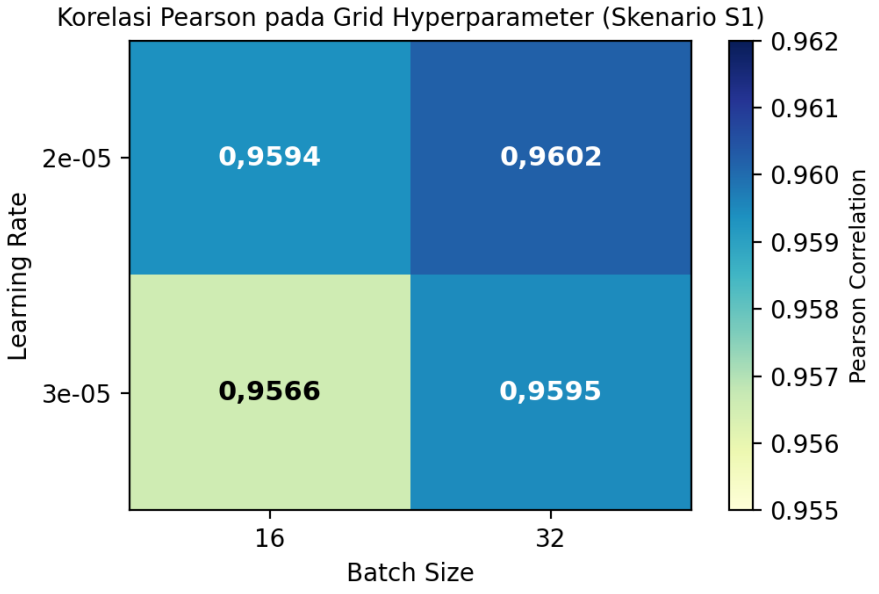


Figure 4: Pearson correlation heatmap across the tested hyperparameter grid for S1.

As shown in the heatmap in Figure 4, the Pearson correlation across all hyperparameter combinations ranged from 0.9566 to 0.9605, a maximum difference of 0.0039. The model's sensitivity to the choice of learning rate and batch size within the tested range is therefore low.

Model performance by similarity zone

Model performance was analyzed within each similarity zone using pooled predictions from all five folds under the best configuration. Table 6 presents four-class accuracy, MAE, and mean bias for the three scenarios.

Table 6: Evaluation results by similarity zone (four-class accuracy).

Scenario	Zone	Support	Accuracy	MAE	Bias
S1	High	1,367	0.9978	0.0092	+0.0017
S1	Medium	1,270	0.6197	0.0726	+0.0639
S1	Low	1,225	0.5698	0.0974	+0.0681
S1	Negative	1,327	0.8041	0.0895	+0.0650
S1mc	High	1,362	0.9941	0.0094	+0.0004
S1mc	Medium	1,250	0.6240	0.0720	+0.0608
S1mc	Low	1,140	0.5465	0.0987	+0.0676
S1mc	Negative	1,313	0.8165	0.0951	+0.0731
S1mca	High	1,361	0.9949	0.0098	+0.0033
S1mca	Medium	1,245	0.6024	0.0736	+0.0628
S1mca	Low	1,128	0.5585	0.0987	+0.0683
S1mca	Negative	1,310	0.8244	0.0928	+0.0672

Overall four-class zone accuracy reached 0.7547 (S1), 0.7560 (S1mc), and 0.7561 (S1mca). The high zone was recognized almost perfectly across all three scenarios (0.9941 to 0.9978), with bias below +0.0033. This ease of recognition is consistent with how the zone was constructed: high-zone pairs originate from paraphrases with the smallest degree of meaning alteration, such that lexical similarity between the two segments remains high and aligned with their semantic similarity, a combination that is most readily recognized by contextual embedding-based models. The negative zone achieved an accuracy of 0.8041 to 0.8244, with bias ranging from +0.0650 to +0.0731.

The medium zone (0.6024 to 0.6240) and low zone (0.5465 to 0.5698) were the two zones with the lowest accuracy. Bias in both zones was positive and of similar magnitude to that of the negative zone, ranging from +0.0608 to +0.0683, as visualized in Figure 6, indicating that the errors are directional rather than random variation. Table 7 details the two dominant error pathways accounting for the majority of classification errors.

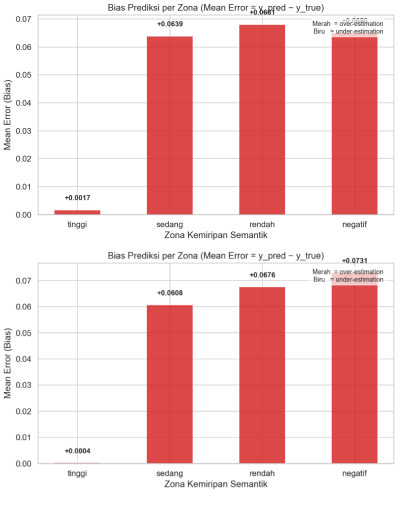
Table 7: Dominant classification error pathways.

Error Pathway	S1	S1mc	S1mca
Medium → High	466 pairs (36.7%)	452 pairs (36.2%)	478 pairs (38.4%)
Low → Medium	410 pairs (33.5%)	403 pairs (35.4%)	394 pairs (34.9%)
Negative → Low	251 pairs (18.9%)	232 pairs (17.7%)	224 pairs (17.1%)

All three error pathways in Table 7 shift one level toward a higher zone, and no opposite-direction error pathway occurs at a comparable proportion (high to medium errors account for only 3 pairs, or 0.2%, in S1). The confusion matrix in Figure 5 confirms this pattern: errors are concentrated along the adjacent upper diagonal rather than being randomly distributed across all classes.



Figure 5: Four-zone confusion matrices for the S1, S1mc, and S1mca scenarios.



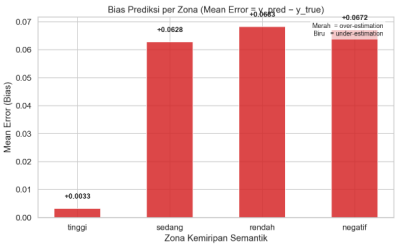


Figure 6: Prediction bias by zone for the S1, S1mc, and S1mca scenarios.

Comparison with internal baselines

Table 8 summarizes the performance of the main model and all comparative methods under the S1 scenario, reported as mean ± standard deviation across the five folds.

Table 8: Comparison of the main model and internal baselines across preprocessing scenarios.

Scenario	Method	Pearson	Spearman	RMSE	MAE
S1	TF-IDF	0.8474 ± 0.0035	0.9441 ± 0.0036	0.3966 ± 0.0048	0.3586 ± 0.0060
S1	Character n-gram Jaccard (3–5)	0.8143 ± 0.0047	0.9081 ± 0.0035	0.3742 ± 0.0061	0.3353 ± 0.0061
S1	Word n-gram Jaccard (1–2)	0.7843 ± 0.0033	0.9212 ± 0.0026	0.4521 ± 0.0040	0.4145 ± 0.0052
S1	IndoBERT without fine-tuning	0.8696 ± 0.0129	0.8780 ± 0.0129	0.2434 ± 0.0028	0.1749 ± 0.0031
S1	Main model (fine-tuned)	0.9602 ± 0.0018	0.9687 ± 0.0018	0.0925 ± 0.0027	0.0661 ± 0.0018
S1mc	TF-IDF	0.8386 ± 0.0047	—	0.4004 ± 0.0062	0.3605 ± 0.0072
S1mc	Character Jaccard (3–5)	0.8065 ± 0.0062	—	0.3755 ± 0.0062	0.3346 ± 0.0062
S1mc	Word Jaccard (1–2)	0.7752 ± 0.0045	—	0.4561 ± 0.0048	0.4166 ± 0.0059
S1mc	IndoBERT without fine-tuning	0.8720 ± 0.0129	—	0.2404 ± 0.0033	0.1714 ± 0.0030
S1mc	Main model (fine-tuned)	0.9602	—	0.0941	0.0672
S1mca	TF-IDF	0.8385 ± 0.0050	—	0.4004 ± 0.0065	0.3605 ± 0.0077
S1mca	Character Jaccard (3–5)	0.8063 ± 0.0065	—	0.3759 ± 0.0065	0.3351 ± 0.0066
S1mca	Word Jaccard (1–2)	0.7749 ± 0.0051	—	0.4569 ± 0.0052	0.4175 ± 0.0063
S1mca	IndoBERT without fine-tuning	0.8713 ± 0.0130	—	0.2399 ± 0.0034	0.1710 ± 0.0029
S1mca	Main model (fine-tuned)	0.9605	—	0.0941	0.0670

The main model decisively outperformed all independent baselines, as summarized in Figure 7. The highest Pearson correlation among the four baselines was achieved by IndoBERT without fine-tuning (0.8696), whereas the main model reached 0.9602, a difference of +0.0903 with a 95% bootstrap confidence interval of [+0.0812, +0.0998]. On RMSE, the main model likewise outperformed the same best-performing baseline (0.0925 versus 0.2434), a difference of −0.1509 [−0.1543, −0.1476]. All differences between the main model and the four baselines were significant across all metrics, as summarized in Figure 8.

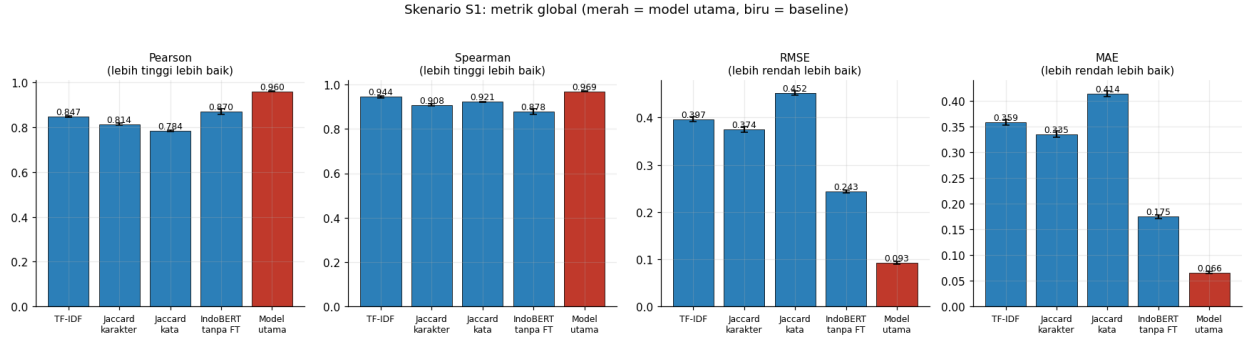


Figure 7: Comparison of global metrics between the main model and internal baselines under S1.

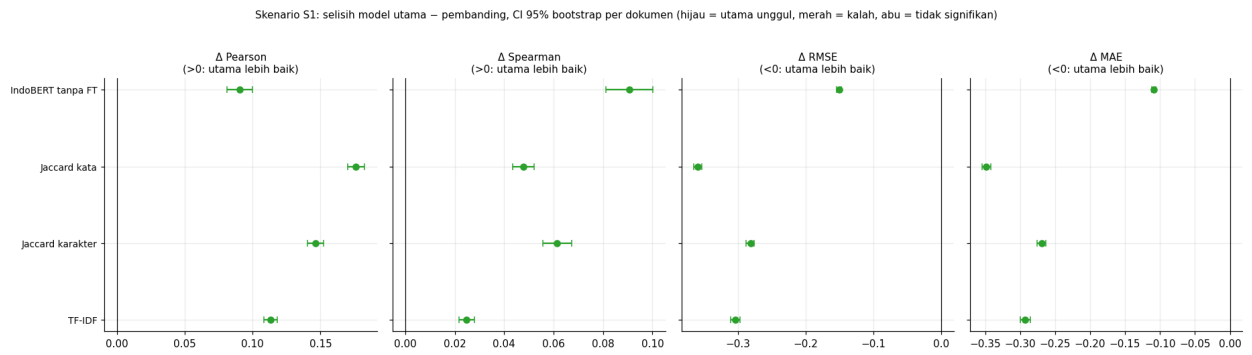


Figure 8: Differences between the main model and baselines with 95% document-level bootstrap confidence intervals under S1.

The performance gap with the baselines was likewise evident at the zone level. The best-performing baseline, IndoBERT without fine-tuning, achieved a four-class (exact-match) zone accuracy of 99.6% in the high zone but declined to 77.5% in the medium zone, 12.7% in the low zone, and only 1.1% in the negative zone (S1 scenario). This pattern confirms that the main model's advantage is not merely a matter of aggregate global regression metrics but holds consistently across the full similarity spectrum: the main model maintained zone accuracy of 80.4 to 82.4% in the negative zone and 54.6 to 57.0% in the low zone (the Results section), far exceeding the baseline in precisely the two zones that are most difficult to distinguish either lexically or through contextual representations without fine-tuning.

In terms of computational efficiency, fine-tuning did not alter the backbone architecture, such that the inference cost of the main model is equivalent to that of IndoBERT without fine-tuning (approximately 28 to 34 ms per pair on an NVIDIA GeForce RTX 3050 Laptop GPU), whereas the lexical baselines (TF-IDF, Jaccard) ran on CPU with inference times below 1 ms per pair. The main model's significant accuracy gains were therefore achieved at an inference computational cost comparable to that of the pretrained encoder alone, rather than through additional architectural overhead.

To confirm that the main model's performance advantage was not specific to a single preprocessing scenario, Table 8 summarizes the same comparison under the S1mc and S1mca scenarios as a robustness check, with the main model's values taken from the best configuration of each scenario in Table 5.

The same pattern held consistently across both additional scenarios: the main model outperformed all independent baselines by a margin comparable to that observed under S1, indicating that the contribution of fine-tuning is robust to the citation- and visual-reference-masking strategy, rather than an artifact of a single preprocessing configuration. The main model's values in this table are not accompanied by RMSE/MAE standard deviations, as they were taken directly from Table 5 and were not recomputed within this baseline-comparison pipeline.

Discussion

Discussion and analysis

The model's error profile was uniform across scenarios. The global mean bias was +0.0488 (S1), +0.0493 (S1mc), and +0.0491 (S1mca), with a global MAE of 0.0661 to 0.0672. Based on Table 6, the high and negative zones were the two zones with the highest accuracy, at 0.9941 to 0.9978 and 0.8041 to 0.8244 respectively, such that model error was concentrated in the two intervening zones.

The performance bottleneck lay in the medium and low zones. The primary cause is geometric in nature: the mean predicted score in the medium zone was 0.8493 (S1), 0.8463 (S1mc), and 0.8481 (S1mca), while the high-zone threshold is set at 0.88. The distance between the mean prediction and this threshold is only 0.0307 to 0.0337, smaller than the medium zone's own MAE (0.0720 to 0.0736). In other words, an error of average magnitude is already sufficient to shift a pair from the medium zone into the high zone. The same pattern holds at the low-to-medium boundary of 0.66, with a mean low-zone prediction of 0.6048 to 0.6090 and MAE of 0.0974 to 0.0987.

The consistent positive bias across all non-high zones relates to the hard-negative sampling strategy used in dataset construction: pairs in the low and negative zones nonetheless originate from the homogeneous Informatics Chapter II domain, retaining substantial lexical and technical-term similarity despite low semantic relatedness. This condition

makes it difficult for the model to separate meaning similarity from surface-level vocabulary similarity, and is more appropriately regarded as a consequence of data characteristics rather than a failure of the model to learn semantic representations (C. Guo et al., 2017). This factor is further reinforced by differences in how pairs were constructed across zones: the high and medium zones were formed through direct paraphrasing of the same segment, whereas the low and negative zones were formed through the selection of different (cross-paragraph) segment pairs without a paraphrasing process. This difference in construction mechanism may introduce additional surface-level signals, such as residual sentence patterns from paraphrasing or structural similarity between segments originating from the same document, which the model may partly learn alongside genuine semantic similarity. Under the current data design, the contribution of such surface-level signals to model performance cannot be measurably separated from the contribution of semantic representation, and this is accordingly noted as a limitation of the dataset design.

Stability across folds was maintained. The range of zone accuracy across folds was 0.7375 to 0.7711 for S1, 0.7331 to 0.8079 for S1mc, and 0.7423 to 0.7804 for S1mca, with per-fold Pearson correlation ranging from 0.9553 to 0.9641 across all scenarios. The weakest fold was not consistent across scenarios (fold 1 for S1 and S1mc, fold 3 for S1mca), indicating that between-fold variation more likely reflects document composition within each fold rather than a systematic weakness in a particular subset. The document-based 5-Fold GroupKFold design ensures that this estimate is free of data leakage between segments of the same document (Kaufman, Rosset, Perlich and Stitelman, 2012).

These findings carry practical implications for the deployment of the model within a plagiarism detection system. Because dominant errors shift one level upward and are concentrated around the 0.88 and 0.66 thresholds, system output within these ranges is not suitable for fully automated decision-making. Two data-supported mitigation options are zone-based threshold calibration to compensate for the +0.06 to +0.07 bias, or the establishment of an uncertainty band around each threshold that is routed to manual review. The resulting continuous score remains usable as a candidate-ranking aid, supported by a Spearman correlation of 0.9683 to 0.9690 across all three scenarios.

Limitations of this study include its single-domain scope (45 Informatics documents from a single institution), the absence of cross-domain or cross-institution testing, and the loss of cross-segment context resulting from the 300-token sliding-window mechanism. In addition, the ground truth was constructed from an automatic labeling model validated against annotators, rather than from full human annotation across all 5,189 pairs.

Conclusion

This study addressed the research question of how a semantic similarity detection system can be implemented and objectively evaluated for Chapter II of Indonesian-language undergraduate theses, through the development of a Fine-Tuned IndoBERT Siamese Bi-Encoder model encompassing document extraction, text segmentation, construction of paired data under three preprocessing scenarios, automatic labeling validated against human annotation (Krippendorff's Alpha of 0.8070 and a label-to-annotator Pearson correlation of 0.9804), and fine-tuning and evaluation through document-based 5-Fold cross-validation. With similarity scores computed directly from cosine similarity without range transformation, the model achieved a Pearson correlation of 0.9602 to 0.9605 and a Spearman correlation of 0.9687 to 0.9690 across the three scenarios, with prediction errors of RMSE 0.0925 to 0.0941 and MAE 0.0661 to 0.0672. The main model consistently outperformed all independent baselines, namely TF-IDF, character n-gram Jaccard, word n-gram Jaccard, and IndoBERT without fine-tuning, with Pearson differences of +0.09 to +0.18, validated as significant through document-level cluster bootstrap at p approaching 0.001. The contribution of fine-tuning to this task is thus empirically confirmed, rather than merely a consequence of the pretrained backbone architecture. The performance difference among the three preprocessing scenarios itself fell below the between-fold standard deviation, with a Pearson difference of only 0.0003 against a standard deviation of 0.0020 to 0.0031, indicating that citation- and visual-reference-masking strategies did not yield a measurable performance improvement on this dataset.

Evaluation across the four similarity zones yielded an overall accuracy of 0.7547 to 0.7561. The model recognized the high and negative zones very well, with accuracies of 0.9941 to 0.9978 and 0.8041 to 0.8244 respectively, but declined sharply in the medium and low zones, with accuracies of 0.6024 to 0.6240 and 0.5465 to 0.5698 respectively. The model's principal limitation lies in a systematic overprediction bias of +0.0608 to +0.0731 across all non-high zones, which consistently produces two dominant error pathways: a medium-to-high zone shift accounting for 36.2 to 38.4 percent of the medium zone's support, and a low-to-medium shift accounting for 33.5 to 35.4 percent. The magnitude of this bias is comparable to the mean distance between predicted scores and the threshold of the zone above, indicating that the model's errors are systematic and directionally predictable rather than random variation.

These findings carry direct implications for the practical deployment of the system. Model output within the ranges surrounding the 0.66 and 0.88 thresholds is not suitable as a basis for automated decision-making without further verification. Three relevant directions for future development to address these limitations are zone-based threshold calibration, the use of a Cross-Encoder as an additional verification stage for candidates falling within the uncertainty band around both thresholds, and the expansion of data coverage beyond a single study program and institution to test the model's generalization across more diverse domains and writing populations.

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Conflict of Interest

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Declaration of AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) to assist with manuscript formatting, structural alignment, and language consistency. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

[Author name]: [CRediT roles to be completed].

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